

MicroMatte 1415-EZ

Introduction

MicroMatte 1415-EZ is a micronized composite wax based on modified synthetic wax. The composite components significantly improve dirt pick up resistance (DPUR), abrasion resistance, burnish resistance, and scratch resistance, while also enhancing slip and lubricity. MicroMatte 1415-EZ can be compared to other popular additives for architectural coatings.

Physical Property:	MicroMatte 1415-EZ	PropylMatte 31HD	PropylMatte 31	AquaMatte 26HD	MicroMatte 1011UVW
Composition	Modified synthetic wax	Densified polypropylene	Polypropylene	Densified oxidized polyethylene	Modified polypropylene
Melting point °C	136 - 140	160 - 170	160 - 170	105 - 111	150 - 156
Density at 25 °C (g/cc)	1.06	1.07	0.89	1.08	1.07
Mean particle size (µm)	10.0 – 15.0	8.0 - 12.0	8.0 - 12.0	6.0 - 8.5	5.0 - 7.5
Maximum particle size (µm)	44.0	31.00	31.00	26.00	22.00

Objective

To evaluate the performance of MicroMatte 1415-EZ in hard urethane and acrylic paint formulation.

Test Methods

- ASTM D4060 Standard Test Method for Abrasion Resistance
- ASTM D3363 Standard Test Method for Film Hardness by Pencil Test (Scratch)
- ASTM D1894 Standard Test Method for Coefficient of Friction
- ASTM D523 Standard Test Method for Gloss Measurement
- ASTM D6736 Standard Test Method for Burnish Resistance
- MPI Soil Release and Dirt Pickup Resistance (DPUR) Study

Sample Preparation

- The coatings for the Taber abrasion test were drawn down with a #60 wire wound rod to produce approximately 2.4 mils (60 µm) DFT coatings.
- The coatings for the scratch resistance test were drawn down with a #28 wire wound rod to produce approximately 1.00 mil (25 microns) DFT coatings.
- For the burnish, gloss and COF tests, coatings were drawn down with a 3-mil bird bar to produce approximately 1.20 mil (30 µm) DFT coatings.
- For Dirt Pick Up Resistance (DPUR) using a Gardco Washability/Wear Tester:

- Made 10 mil (254 μm) wet film thickness drawdowns of the paint; dried 7 days
- Placed 3 drops of brown iron oxide dispersion on the coating
- Spread the dispersion using a sponge brush; dried 3 days
- Affixed drawdown to tester. Placed 2 ml of water on the coating and ran for 1 cycle
- Repeat (d) for 9 more cycles
- Ran 20 additional cycles without further water addition
- After 30 cycles, removed the panel, and rinsed off stained area with water
- After the panels have dried, took the final color reading using spectrophotometer
- Measured delta E multiple times (at least 5) across the panel and averaged the results.

- All coatings are allowed to cure for 7 days at room temperature at ~50% humidity.

Formulations

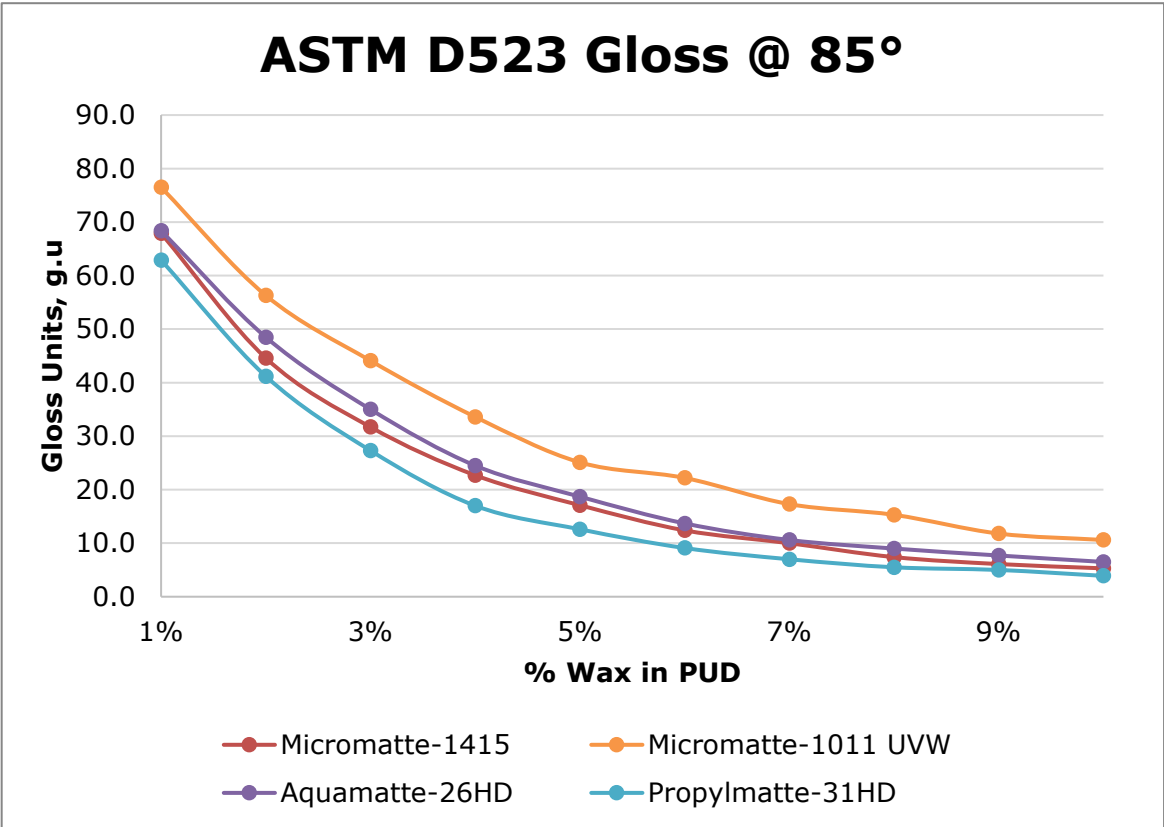
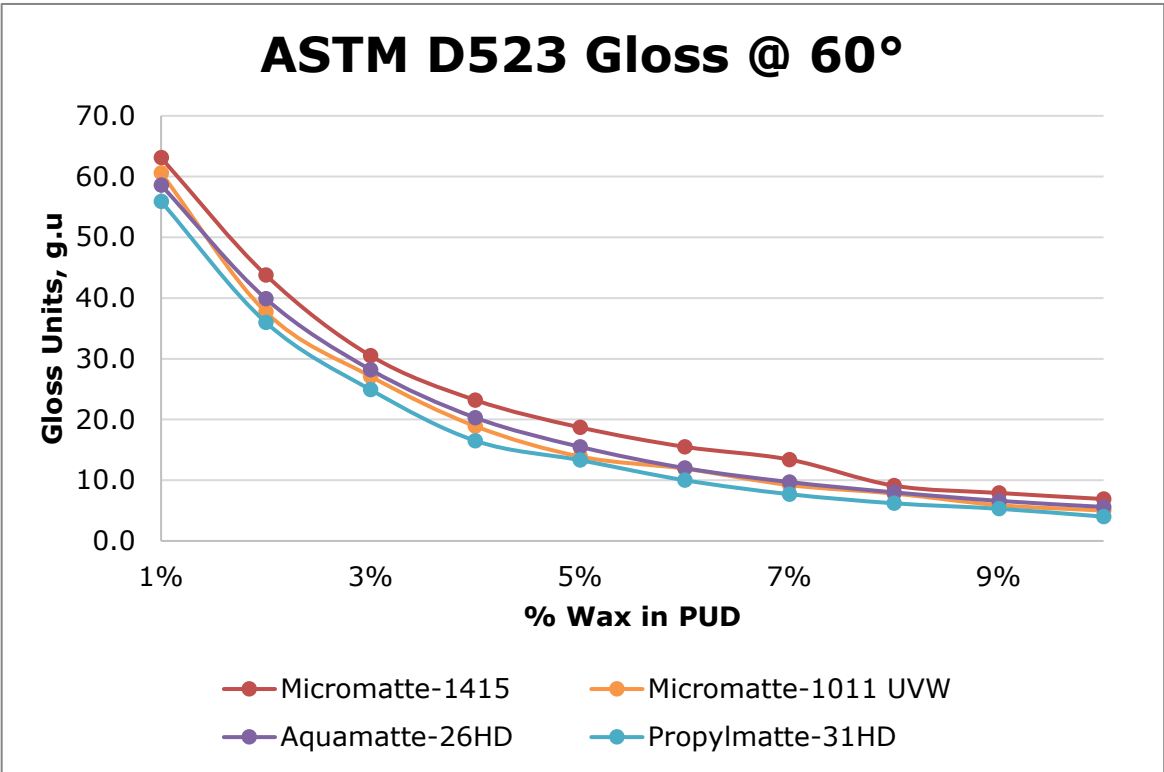
All waxes in the study were incorporated in the grind phase of the formulations.

Hard Urethane Formula:	Weight %:
UROTUF®L63-MPW-38 (Reichhold)	97.5
Diethylene glycol monoethyl ether	2.0
BYK-345	0.2
TEGO® Foamex 805 N	0.3
Total:	100.0
Non-volatile (Solids):	38.4

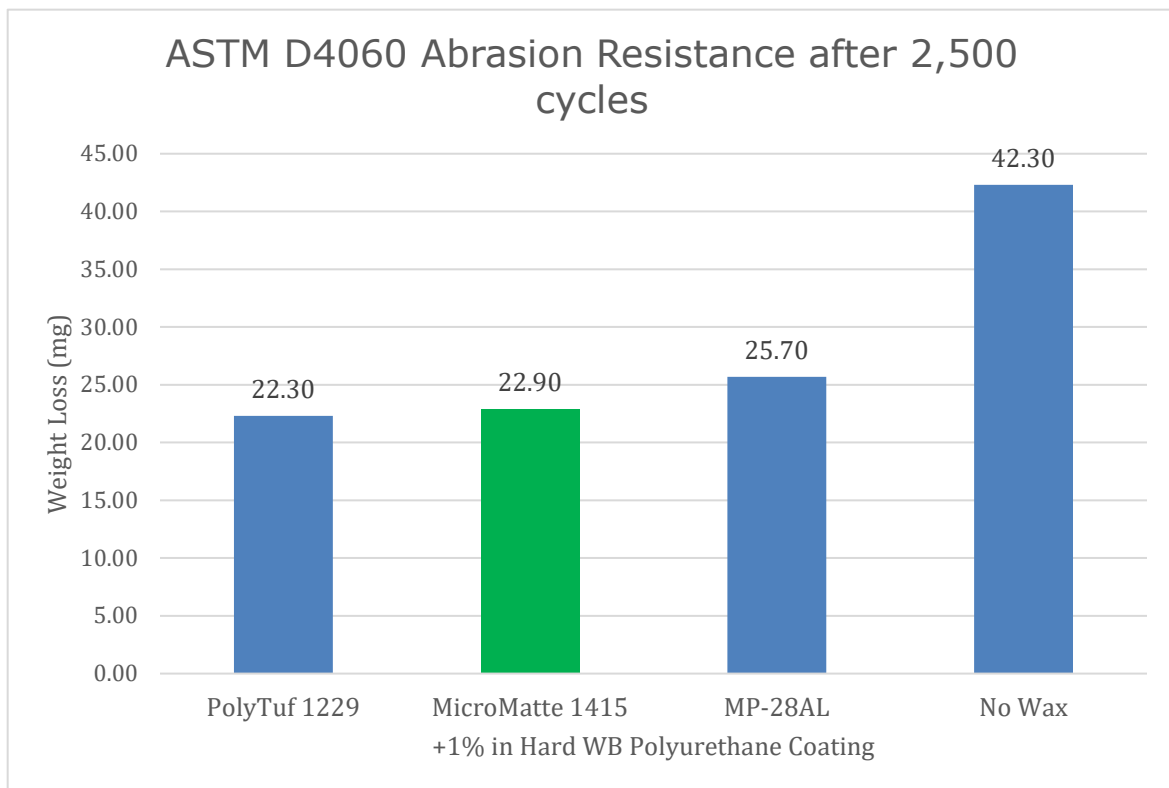
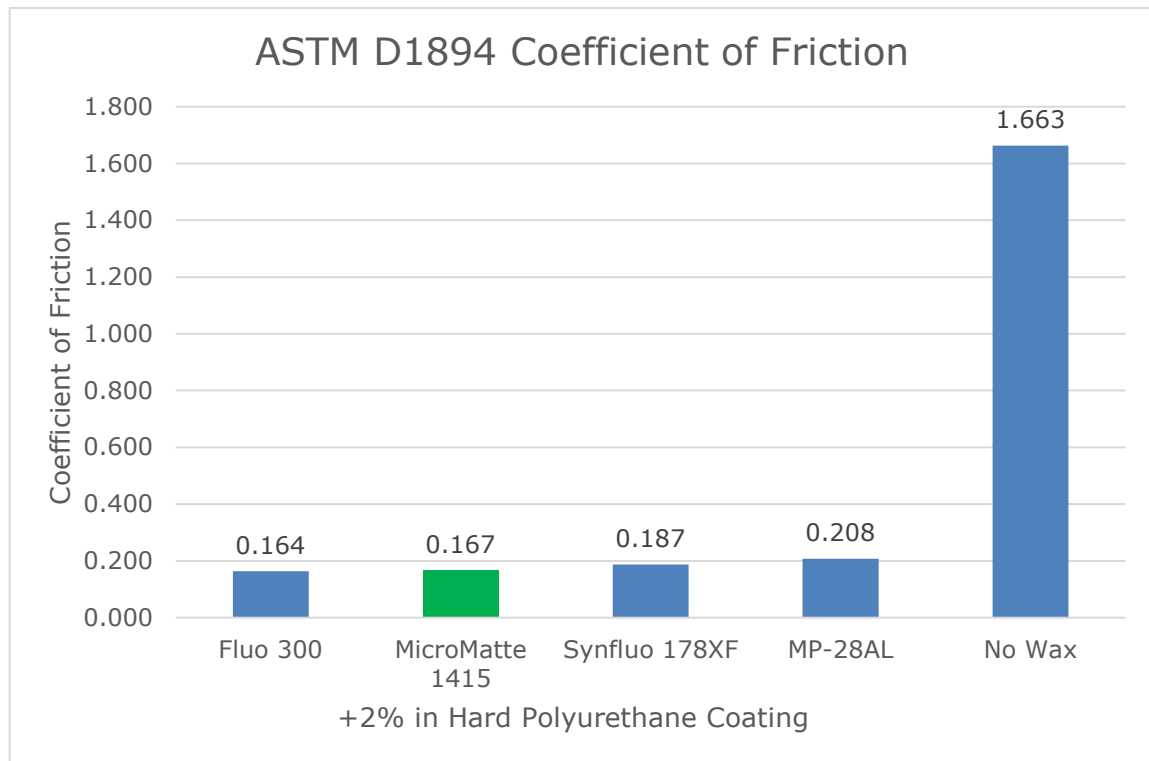
MPI Water-Based Acrylic Paint Formula:	Weight %:
Water	24.09
Natrosol Plus 330 PA	0.20
BYK-348	0.20
Propylene Glycol	1.03
Dowsil 69	0.11
Black AIT 744 (42% Lampblack)	4.80
Minex 4	30
Xanthan gum	0.10
Proxel BZ	0.16
Carbowet GA-211	0.16
Wax	2.00
Rhoplex VSR-2015 Binder (50% solids)	35
Acrysol RM-2020NPR	2.15

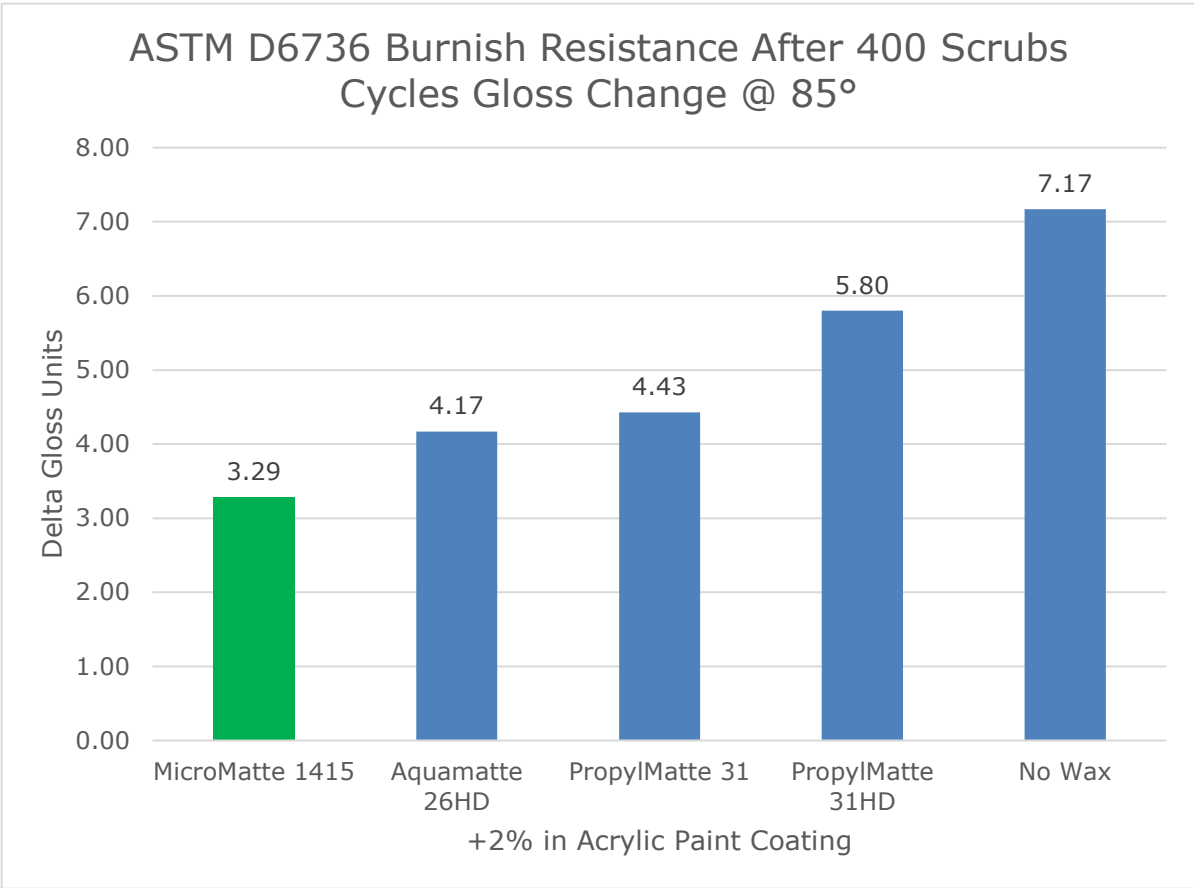
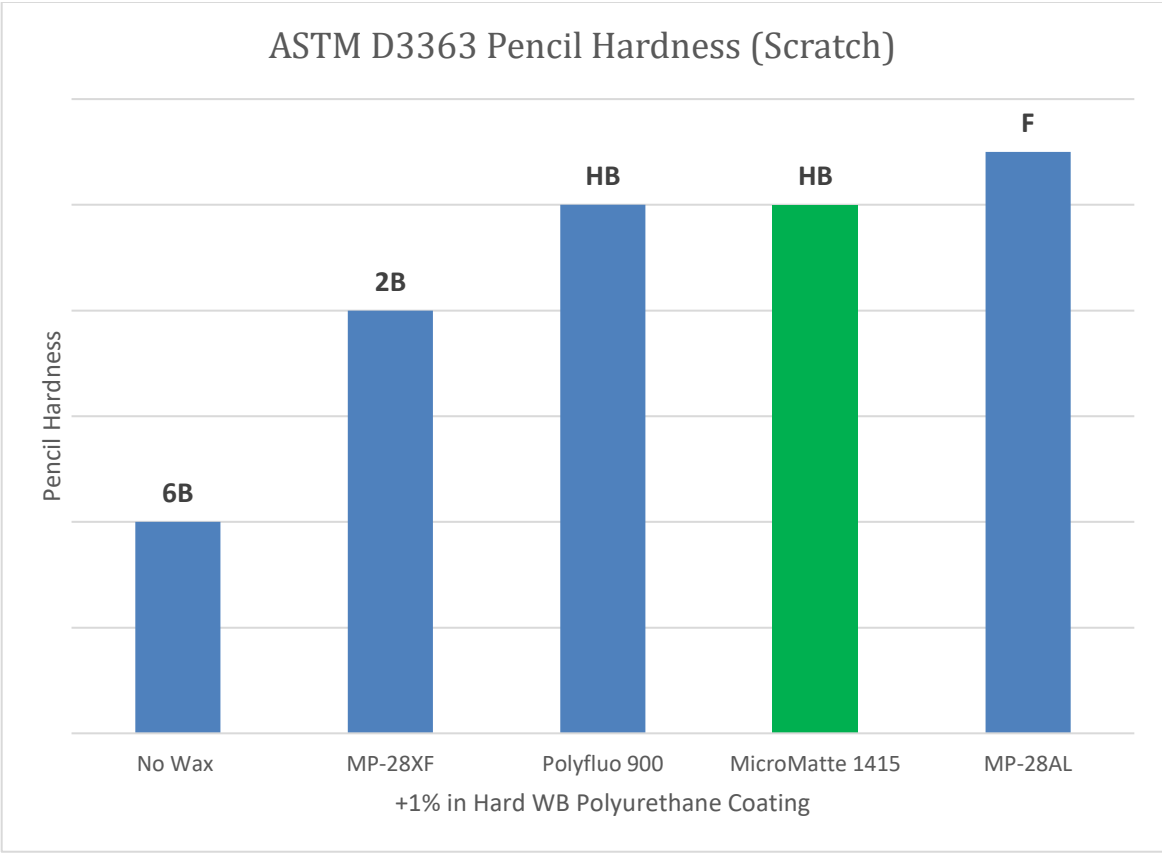
Brown Iron Oxide Dispersion:	Weight %:
Yellow iron oxide dispersion in glycerin	33.33
Red iron oxide dispersion in glycerin	33.33
Black iron oxide dispersion in glycerin	33.33

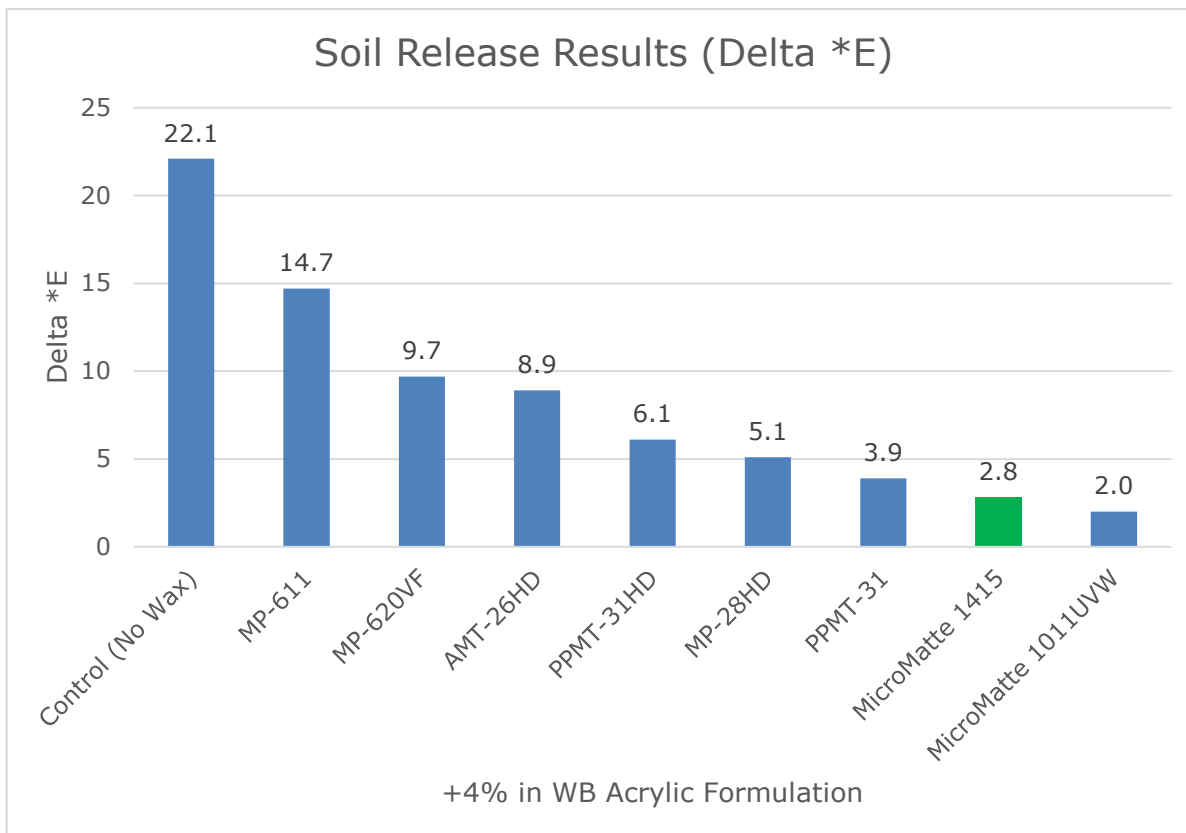
Results



For COF, abrasion, and scratch hardness testing, MicroMatte 1415-EZ was compared vs. other key MPI products known for superior lubricity and surface durability







Conclusions

- Gloss reduction (60°)**

MicroMatte 1415-EZ shows similar but slightly less efficient gloss reduction compared to PropylMatte 31 and AquaMatte 26HD, with PropylMatte 31HD performing best.

- Gloss reduction (85°)**

MicroMatte 1415-EZ shows comparable gloss reduction vs. AquaMatte 26HD, with PropylMatte 31HD performing best.

- **Coefficient of Friction**

MicroMatte 1415-EZ has excellent slip and lubricity with a kinetic COF of 0.164. It has a kinetic COF almost identical to a PTFE powder such as Fluo 300.

- **Abrasion resistance**

MicroMatte 1415-EZ has the second-best abrasion data of all Micro Powders fine powder products with a 22.9 mg weight loss after 2,500 cycles.

- **Scratch resistance**

MicroMatte 1415-EZ shows pencil hardness properties similar to other key MPI products known to provide excellent scratch resistance.

- **Burnish resistance**

MicroMatte 1415-EZ has the best resistance to burnishing (polishing) of any MPI product. It shows superior burnish resistance (gloss-up after 400 scrub cycles) compared to the other products, with only a 3.29 gloss unit decrease at 60°.

- **Dirt Pick Up Resistance (Soil release)**

MicroMatte 1415-EZ has superior dirt pick up resistance (DPUR) and is very similar in performance to MicroMatte 1011 UVW, with a Delta E of 2.8 after staining and cleaning